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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
Regular EXAMINATION October 2026
Fifth SEMESTER
Discipline Specific Elective (DSE) - MG5DSECHEA01 - FORENSIC CHEMISTRY
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A
Multiple Choice Questions

Answer all questions. Each question carries 1 marks

1. Which statement about explosive classification is correct?
d. One substance can be described using more than one classification scheme

2. Which cannabis product is specifically derived from the resin of the cannabis plant? a. Charas

3. Toxicology is the study of: b. Poisons and their effects

4. The approximate pH range over which phenolphthalein exhibits its characteristic pink colour is: c. 8.2-10

5. Chain of custody is important because it: b. Establishes documented continuity of evidence

6. Which of the following can cause an accidental fire? a. Defective electrical wiring

7. Ethanol in alcoholic beverages is mainly produced by: b. Fermentation

8. Which poison commonly produces convulsions? a. Strychnine

9. The most sensitive multi-element technique, also capable of isotope-ratio fingerprinting of heavy metals, is:
b. ICP-MS

10. Why should adhesive tape be examined as a layered material? c. Its backing and adhesive may provide different comparison features

11. Fires involving cooking oils and fats are classified as: d. Class F

Part B
Short Answer Questions

Answer any 8 questions. Each question carries 3 marks

12. Define adulteration and give one general beverage example.
Adulteration is the deliberate or accidental addition of inferior, harmful, or unauthorized substances to a product, reducing its quality, purity, or safety. It may mislead consumers and pose health risks. Example: Adding water to milk or diluting fruit juice with water without proper declaration.

13. Explain the role of forensic toxicology in determining the cause and manner of death.
Forensic toxicology detects and measures drugs, alcohol, poisons, and other toxic substances in biological samples. It helps determine whether a substance contributed to or caused death by interpreting its concentration and toxic effects. It assists in determining the manner of death, such as accidental, suicidal, or homicidal, along with other investigative findings.

14. Classify dyes by chemical structure, with an example of each.
Dyes can be classified according to their chemical structure: Azo dyes: Contain an –N=N– group. Example: Methyl orange. Triphenylmethane dyes: Contain a triphenylmethane-based structure. Example: Malachite green.
Anthraquinone dyes: Based on an anthraquinone structure. Example: Alizarin. Indigoid dyes: Based on an indigo-type structure. Example: Indigo.

15. Write a short note on chromatography in forensic toxicology.
Chromatography is a separation technique used to separate components of a mixture based on their different interactions with stationary and mobile phases. It helps detect and identify drugs, poisons, and their metabolites in biological samples such as blood and urine. Common techniques include thin-layer chromatography (TLC), gas chromatography (GC), and high-performance liquid chromatography (HPLC).

16. State the basic roles of gas chromatography and mass spectrometry in GC MS.
Gas chromatography (GC): Separates the volatile components of a mixture. Mass spectrometry (MS): Identifies compounds by analysing their mass-to-charge ratios and fragmentation patterns. GC-MS: Combines separation and identification, enabling sensitive and reliable forensic analysis of drugs, toxins, and other trace substances.

17. What are synthetic drugs?
Synthetic drugs are substances produced through chemical synthesis rather than being obtained directly from natural sources. They may imitate or alter the effects of naturally occurring drugs.
Examples: Amphetamine, methamphetamine, and MDMA (ecstasy).

18. Explain the procedure for collecting stomach contents during autopsy.
Open the stomach carefully during autopsy and examine its contents, noting the appearance, odour, and approximate quantity. Collect the stomach contents in a clean, leak-proof, properly labelled container; preserve a representative sample according

to laboratory protocol. Seal the container, document the collection, and maintain the chain of custody during storage and transport to the forensic laboratory.

19. Define Rf and explain its use in ink comparison.

Rf (retention factor) is the ratio of the distance travelled by a substance to the distance travelled by the solvent front from the origin. Rf = Distance travelled by the substance/Distance travelled by the solvent front.

In ink comparison, TLC is used to separate ink dyes. Comparing their Rf values under identical conditions helps assess whether inks have similar dye components.

20. Mention four presumptive tests used in forensic drug analysis.

Marquis test: Used as a preliminary test for substances such as opiates and amphetamine-type drugs. Scott test: A presumptive test commonly used for cocaine. Duquenois–Levine test: Traditionally used as a presumptive test for cannabis. Dille–Koppanyi test: Used as a presumptive test for barbiturates.

21. What is secondary blast injury?

Secondary blast injury is injury caused by fragments and debris propelled by an explosion. These fragments may originate from the explosive device, surrounding objects, glass, or building materials. Examples: Penetrating wounds, lacerations, and eye injuries caused by flying debris.

Part C
Short Essay Questions

Answer any 3 questions. Each question carries 5 marks

22. Write a short account of cocaine and amphetamine type stimulants.

Cocaine is a naturally occurring stimulant obtained from the leaves of the coca plant. It stimulates the central nervous system by increasing the availability of neurotransmitters, particularly dopamine. Effects include euphoria, increased alertness, raised heart rate, and increased blood pressure. Toxic effects include seizures, cardiac arrhythmias, and heart attack.

Amphetamine-type stimulants are synthetic or semi-synthetic central nervous system stimulants. Examples include amphetamine, methamphetamine, and MDMA (ecstasy). They increase the activity of neurotransmitters such as dopamine and noradrenaline; MDMA also strongly affects serotonin. Adverse effects include insomnia, agitation, hyperthermia, hypertension, and dependence.

23. Describe the emergency care measures provided to patients with acute poisoning.

Ensure safety - Protect the patient and medical personnel from further exposure. Identify the suspected poison, route of exposure, and time of exposure, if possible.

Assess and stabilise vital functions - Follow the ABC approach: airway, breathing, and circulation. Provide oxygen, assisted ventilation, or CPR when clinically indicated.

Prevent further absorption - Remove contaminated clothing and wash exposed skin or eyes with appropriate amounts of water. Do not induce vomiting routinely. Gastrointestinal decontamination, such as activated charcoal, should be considered only when clinically appropriate and under medical guidance.

Administer specific treatment - Give an appropriate antidote when indicated, such as naloxone for suspected opioid-induced respiratory depression. Treat seizures, shock, abnormal temperature, and other complications.

Monitor and investigate - Monitor vital signs, consciousness, ECG, and relevant laboratory parameters. Contact a poison information centre or clinical toxicologist and arrange hospital observation or admission as needed.

24. Explain addition and condensation polymerisation with mechanism and forensic examples.

Addition polymerization - A process in which unsaturated monomers combine to form a polymer without eliminating small molecules. Mechanism: Commonly proceeds through initiation, propagation, and termination in chain-growth polymerisation. Example: Ethylene forms polyethylene. $n\ CH_2{=}CH_2 \rightarrow [-CH_2-CH_2-]n$

Forensic examples: Polyethylene packaging, polypropylene fibres, and acrylic plastic fragments recovered from crime scenes.

Condensation polymerization - A process in which monomers containing suitable functional groups combine, commonly with the elimination of small molecules such as water or methanol. Mechanism: Functional groups react stepwise to form new covalent bonds and progressively longer polymer chains.

Example: Formation of a polyester from a diol and a dicarboxylic acid, with elimination of water.

Diol+Dicarboxylic acid→Polyester+Water

Forensic examples: Polyester textile fibres and nylon fibres recovered from clothing, questioned fabrics, or crime scenes.

25. Explain the collection, preservation, and packing of fire debris.

Collection - Document the fire scene with photographs, notes, and a sketch before collecting samples. Collect representative debris from suspected points of origin, including materials that may contain ignitable-liquid residues. Collect a control sample from an unaffected area for comparison.

Preservation - Minimise handling and exposure to air to reduce the loss of volatile compounds. Use clean tools and avoid cross-contamination between samples. Collect samples promptly and follow laboratory protocols for volatile-residue evidence.

Packing - Place debris in clean, airtight, vapour-resistant containers, such as approved metal cans or suitable glass containers with tight closures. Avoid ordinary plastic bags because volatile residues may escape or interact with the packaging. Seal each container securely and label it with the case number, sample description, location, date, time, and collector's details.
Documentation and transport - Maintain a complete chain-of-custody record. Store and transport samples under appropriate conditions, away from heat and ignition sources. Submit samples to the forensic laboratory for examination, commonly by headspace analysis and GC-MS.
Forensic significance - Fire debris analysis can help detect and identify ignitable-liquid residues and assist in evaluating the possible origin and cause of a fire.

26. Explain sensitivity and stability of explosives.

Sensitivity of explosives - Sensitivity is the degree to which an explosive responds to an external stimulus and initiates a reaction or detonation. Impact sensitivity: Response to mechanical impact. Friction sensitivity: Response to frictional force. Thermal sensitivity: Response to heat or elevated temperature.
Importance: Sensitivity influences the hazards associated with handling, transport, storage, and accidental initiation.
Stability of explosives - Stability is the ability of an explosive to retain its chemical and physical properties during storage and handling.
Chemical stability: Resistance to decomposition over time. Thermal stability: Ability to resist heat-induced decomposition. Environmental stability: Resistance to deterioration due to moisture, light, and other environmental conditions.
Importance: Stable explosives are generally safer to store under approved conditions, while deterioration may increase hazards or reduce performance.
Relationship between sensitivity and stability - Sensitivity describes how readily an explosive initiates in response to a stimulus. Stability describes how well it resists unwanted change or decomposition over time. They are distinct properties: an explosive may be relatively stable during proper storage yet sensitive to a particular initiating stimulus.
Forensic importance: These properties help forensic experts assess explosive hazards, examine damaged or recovered materials, and interpret evidence from explosion scenes.

Part D
Essay Questions

Answer any 2 questions. Each question carries 10 marks

27. Explain how pharmacological, chemical and legal classifications complement one another. Use examples without treating legal categories as permanent or universal.

Pharmacological classification: Classifies substances according to their effects on the body. Example: morphine is an opioid analgesic. Chemical classification: Classifies substances according to their chemical structure or composition. Example: ethanol belongs to the alcohol group. Legal classification: Classifies substances according to laws regulating their possession, sale, and use. Example: morphine is a controlled substance in many countries.
Pharmacological classification explains effects, chemical classification helps identify substances, and legal classification determines regulatory status. All three classifications provide different but related information. Legal categories vary between countries and may change over time.

28. Describe the general management of poisoning. Explain the emergency care of poisoned patients, including stabilization, decontamination, supportive care, and enhanced elimination of poisons.

Initial assessment: Identify the suspected poison, route of exposure, and time of poisoning. Stabilization: Maintain airway, breathing, and circulation using the ABCDE approach. Decontamination: Remove contaminated clothing, wash exposed skin, irrigate eyes, and use activated charcoal when appropriate. Do not induce vomiting. Supportive care: Monitor vital signs and treat seizures, shock, respiratory failure, and metabolic disturbances. Antidotes: Administer specific antidotes when indicated, such as naloxone for opioid poisoning and N-acetylcysteine for paracetamol poisoning. Enhanced elimination: Use selected methods such as urinary alkalinisation, multiple-dose activated charcoal, or haemodialysis when appropriate.
Early stabilization, appropriate treatment, and continuous monitoring are essential for successful management.

29. Compare composition based dating information with aging based measurements. Explain what each can contribute and why neither always provides a precise writing date.

Composition-based dating: Examines ink ingredients, such as dyes, pigments, solvents, and resins, to identify the formulation and establish possible chronological limits. Aging-based measurements: Examine changes in ink over time, such as solvent evaporation, oxidation, and chemical degradation. Methods used: Chemical analysis, including GC–MS, may help examine ink composition or selected aging-related changes. Comparison: Composition-based analysis studies what the ink contains, whereas aging-based analysis studies how the ink changes over time. Limitations: Ink formulations may be used for years, while environmental conditions such as heat, humidity, and light affect aging. Neither method always establishes the exact writing date. Results must be interpreted with other forensic evidence and the limitations of the analytical method.